<FITRA>

RISK MANAGEMENT PLAN

Version *<1.1>*

<04/11/2026>

VERSION HISTORY

Version #	Implemented By	Revision Date (dd/mm/yy)	Approved By	Approval Date (dd/mm/yy)	Reason
0.1	*Jing Jie, Gary Quah, Matthew Wong, Rusyaidi, Yew Yi, Zheng Wee*	*24/09/26*	*Jing Jie*	*24/09/26*	Initial Risk Management Plan draft
0.2	*Jing Jie, Gary Quah, Matthe Wong, Rusyaidi, Yew Yi, Zheng Wee*	*30/09/26*	*Jing Jie*	*30/09/26*	Risk Management Plan edits
0.3	*Jing Jie, Gary Quah, Matthe Wong, Rusyaidi, Yew Yi, Zheng Wee*	*07/10/26*	*Jing Jie*	*07/10/26*	Risk Management Plan edits
0.4	*Jing Jie, Gary Quah, Matthe Wong, Rusyaidi, Yew Yi, Zheng Wee*	*11/10/26*	*Jing Jie*	*11/10/26*	Risk Management Plan edits
1.0	*Jing Jie, Gary Quah, Matthe Wong, Rusyaidi, Yew Yi, Zheng Wee*	*14/10/26*	*Jing Jie*	*14/10/26*	Finalisation of Risk Management plan
1.1	Gary, Yew Yi	29/09/26	Gary	29/09/26	Aligned the risk register (2.3.1) with the Top 10 Risk List (2.5); recorded impact, priority and owner for each risk.

UP Template Version: 11/30/06

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1 INTRODUCTION

1.1 PURPOSE OF THE RISK MANAGEMENT PLAN

The purpose of the risk management plan is to establish a systematic approach to identifying, analysing, monitoring and responding to potential project risks during the development lifecycle. The proposed risk management plan seeks to anticipate and avoid any potential problems through actionable mitigation steps and outline how risk management activities will be recorded and monitored throughout the project. This ensures that the development meets the intended schedule deadlines and deliverables without compromises.

A risk is an event or condition that, if it occurs, could have a positive or negative effect on a project's objectives. Risk Management is the process of identifying, assessing, responding to, monitoring, and reporting risks. This Risk Management Plan defines how risks associated with the Fitra project will be identified, analyzed, and managed. It outlines how risk management activities will be performed, recorded, and monitored throughout the lifecycle of the project and provides templates and practices for recording and prioritizing risks.

The Risk Management Plan is created by the project manager in the Planning Phase of the CDC Unified Process and is monitored and updated throughout the project.

The intended audience of this document is the project team, project sponsor and management.

2 RISK MANAGEMENT PROCEDURE

2.1 PROCESS

The project manager, along with team members, will ensure that risks are actively and thoroughly identified throughout the life cycle of this project. Risks will be identified as early as possible in order to minimise any negative impacts and capitalise on enhancements. The Quality Assurance manager shall serve as the Risk Manager for this project.

2.2 RISK IDENTIFICATION

Identifying risks is the process of understanding what potential events might hurt or enhance this project. This project uses brainstorming to identify these risks.

Risk identification will involve the project team and will include an evaluation of environmental factors, organisational culture and the project management plan, including the project scope. Careful attention will be given to the project deliverables, assumptions, constraints, WBS, cost/effort estimates, resource plan, and other key project documents.

A Risk Management Log will be generated and updated as needed and will be stored electronically in the project library located at ./RiskLogs.

Category	Risks
Manpower	● Developers might not have aligned schedules, causing friction in planning available dates for integration and collaboration.
Budgeting	● API usage might exceed the planned limits, resulting in inaccessible APIs stalling development.
Technological	● Hardware required to develop the application might fail, such as desktops and laptops. ● APIs might go down due to maintenance or migration of API versioning/access methods, resulting in delays in development.
Development	● Dependencies on other developers might arise for integration of features. ● Development might stall due to stagnant pull requests waiting to be approved. ● Lack of secure coding practices might lead to security debts ● Lack of code reviews might result in tech debt and quality issues. ● Lack of testing during development might result in bugs or code smells delaying development/production.
Documentation	● Documentation might cause delays in development due to documentation debt.

2.3 RISK ANALYSIS

All risks identified will be assessed to identify the range of possible project outcomes. Qualitative analysis will be used to determine which risks are the top risks to pursue and respond to and which risks can be ignored.

2.3.1 Qualitative Risk Analysis

The probability and impact of occurrence for each identified risk will be assessed by the project manager, with input from the project team using the following approach:

Probability

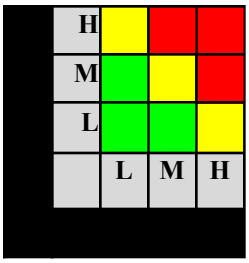
● High – Greater than *70%* probability of occurrence

● Medium – Between *30%* and *70%* probability of occurrence

- Low – Below *30%* probability of occurrence

No.	Risk event	Description	Probability %	Impact	Priority (zone)	Owner
1	**Development deficit**	Development might face roadblocks due to dependencies on other developers	75%	High	Red	Matthew Wong
2	**Documentation deficit**	Documentation might cause delays in development due to documentation debt.	75%	High	Red	Gary Quah
3	**Quality Assurance**	Quality assurance might cause delays in development due to under- or excessive testing.	60%	Low	Green	–
4	**Time & Schedule management**	Team members have different schedules and levels of commitment to the module, causing friction in deadlines.	60%	High	Red	Gary Quah
5	**Quality Assurance testing**	Lack of active testing during development might creep up, causing tech debt.	50%	High	Red	Rusyaidi
6	**Version Control**	Pull requests might take time to approve; merge conflicts might arise, causing delays in development.	40%	Medium	Yellow	Matthew Wong
7	**Availability of API**	APIs might not be readily available during development due to maintenance or unavailability.	10%	High	Yellow	Yew Yi
8	**Security issues**	Security issues might arise during development, such as exposure of API keys.	10%	High	Yellow	Zheng Wee
9	**Technological failure**	Hardware used to develop the application, such as desktops or laptops, might fail.	10%	Low	Green	–
10	**Budgeting**	Exceeding the budget for paid API tokens might cause blockers in development due to inaccessible APIs until payment.	10%	Medium	Green	–

11	**Scope creep**	Attempting to overbuild features for the application might cause delays in development.	40%	Medium	Yellow	Jing Jie
12	**Communication & Asynchronous coordination breakdowns**	Misunderstanding & miscommunication of requirements and features.	40%	Medium	Yellow	Yew Yi



Impact

- High – Risk that has the potential to greatly impact project cost, project schedule or performance
- Medium – Risk that has the potential to slightly impact project cost, project schedule or performance
- Low – Risk that has relatively little impact on cost, schedule or performance

The priority of each risk in the table above is the zone given by its probability and impact on this matrix. Risks that fall within the RED and YELLOW zones will have risk response planning which may include both a risk mitigation and a risk contingency plan.

2.4 RISK RESPONSE PLANNING

Each major risk (those falling in the Red & Yellow zones) will be assigned to a project team member for monitoring purposes to ensure that the risk will not "fall through the cracks". The assigned owner of each risk is named in the Owner column of the table in Section 2.3.1.

For each major risk, one of the following approaches will be selected to address it:

- **Avoid** – eliminate the threat by eliminating the cause
- **Mitigate** – Identify ways to reduce the probability or the impact of the risk
- **Accept** – Nothing will be done
- **Transfer** – Make another party responsible for the risk (buy insurance, outsourcing, etc.)

For each risk that will be mitigated, the project team will identify ways to prevent the risk from occurring or reduce its impact or probability of occurring. This may include prototyping, adding tasks to the project schedule, adding resources, etc.

For each major risk that is to be mitigated or that is accepted, a course of action will be outlined for the event that the risk does materialize in order to minimize its impact.

2.5 RISK MONITORING, CONTROLLING, AND REPORTING

The level of risk on a project will be tracked, monitored and reported throughout the project lifecycle.

A "Top 10 Risk List" will be maintained by the project team and will be reported as a component of the project status reporting process for this project. . The list is taken from the risks in Section 2.3.1, ranked by priority zone (Red, then Yellow, then Green), then by impact, then by probability. Quality Assurance (No. 3) and Technological failure (No. 9) are Green and rank 11th and 12th, so they are not in the Top 10; they remain in Section 2.3.1 and are reviewed with the Risk Log (Section 3).

All project change requests will be analysed for their possible impact to the project risks.

Management will be notified of important changes to risk status as a component to the Executive Project Status Report.

Risk event	Description	Rank this month	Rank last month	Number of months in Top 10	Risk Resolution Process
Development deficit	Development might face roadblocks due to dependencies on other developers	1	1	1	Implement development priority; ensure highly dependent features are completed in advance
Documentation deficit	Documentation might cause delays in development due to documentation debt.	2	2	1	The project manager is to actively check on the documentation progress to ensure documentation does not stagnate and creep up on the team.
Quality	Insufficient testing might	3	3	1	End-to-end testing

assurance	cause bugs in features.				must be done to test affected components of code prior to creating any merges into the development / main branch.
Time & Schedule management	Team members have different schedules and levels of commitment to the module causing friction in deadlines.	4	4	1	Team members have fixed dates and times to have regular meetings each week on Wednesday and Sunday.
Availability of third party APIs	APIs might not be readily available during development due to maintenance or unavailability.	5	5	1	Mock data will be prepared to simulate API calls in the event of API failure.
Version Control	Pull requests might take time to approve; merge conflicts might arise, causing delays in development.	6	6	1	Team members will have active code reviews for pull requests to ensure pull requests do not stack up, facilitated by the lead developer.
Security issues	Security issues might arise during development, such as exposure of API keys.	7	7	1	Team members are to apply the fundamentals of application security such as security headers and non-public API keys stored in an environment file or vault.
Scope creep	Attempting to overbuild features for the application might cause delays in development.	8	8	1	Team members are to strictly adhere to the functional requirements defined in the SRS (Software Requirements Documentation)

Communication & Asynchronous coordination breakdowns	Misunderstanding & miscommunication of requirements and features.	9	9	1	Active communication shall be carried out in Microsoft Teams to ensure no breakdown in communication occurs.
Budgeting	Exceeding the budget for paid API tokens might cause blockers in development.	10	10	1	Active monitoring of token usage will be done through individual API dashboards.

3 TOOLS AND PRACTICES

A Risk Log will be maintained by the project manager and quality manager, and will be reviewed as a standing agenda item for project team meetings.

RISK MANAGEMENT PLAN APPROVAL

The undersigned acknowledge they have reviewed the **Risk Management Plan** for the Fitra project. Changes to this Risk Management Plan will be coordinated with and approved by the undersigned or their designated representatives.

Signature:	JJ	Date:	30/09/2026
Print Name:	Jing Jie		
Title:	QA Manager		
Role:			

Signature:	GQ	Date:	30/09/2026
Print Name:	Gary Quah		
Title:	Project Manager		
Role:			

Signature:	ZW	Date:	30/09/2026
Print Name:	Zheng Wee		
Title:	QA Engineer		
Role:			

Signature:	RS	Date:	30/09/2026
Print Name:	Rusyaidi		
Title:	QA Engineer		
Role:			

Signature:	YY	Date:	30/09/2026
Print Name:	Yew Yi		
Title:	Release Engineer / Manager		
Role:			

Signature:	MW	Date:	30/09/2026
Print Name:	Matthew		

Title:	Lead Developer
Role:	

APPENDIX A: REFERENCES

The following table summarizes the documents referenced in this document.

Document Name and Version	Description	Location
Software Requirements Specification (SRS) v1.0	Defines the functional and non-functional requirements and scope for the Fitra application.	o/System_documentation/Lab02/SC3040_Lab02_SoftwareRequirementSpecifications.pdf

APPENDIX B: KEY TERMS

The following table provides definitions for terms relevant to the RISK MANAGEMENT PLAN.

Term	Definition
Application Programming Interface (API)	A set of rules, protocols or endpoints allowing the Fitra application to communicate with external third party services and data providers.
Risk	An event or condition that, if occurs has a positive or negative impact on the project.
Risk mitigation	A response strategy aimed at reducing the probability of occurrence of a risk.
Scope creep	Uncontrolled changes or continuous growth in a project's scope beyond the defined functional requirements resulting in insufficient time or resources to complete the defined scope.
Technical / documentation debt	The cost of additional rework caused by poor or underperforming code or documentation.